

Linear Transformation

Linear Algebra

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Transformations / Functions / Mapping.

Linear Transⁿ.

Linear Transformation

Definition (Linear Transformation)

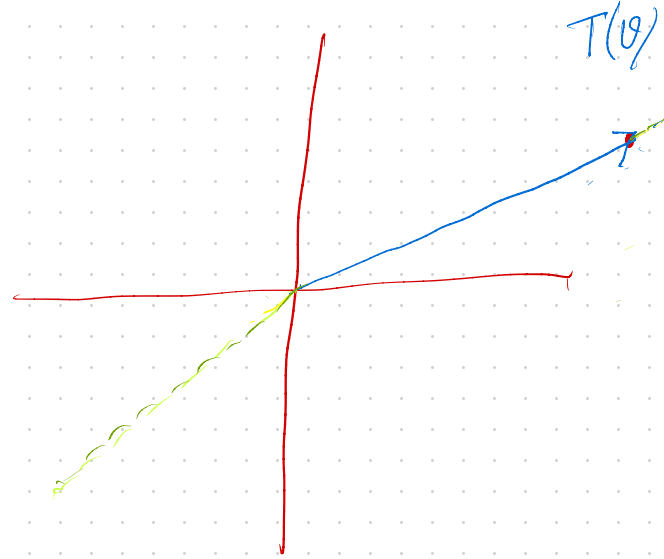
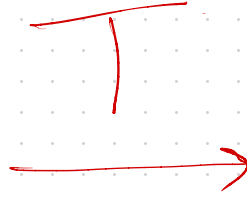
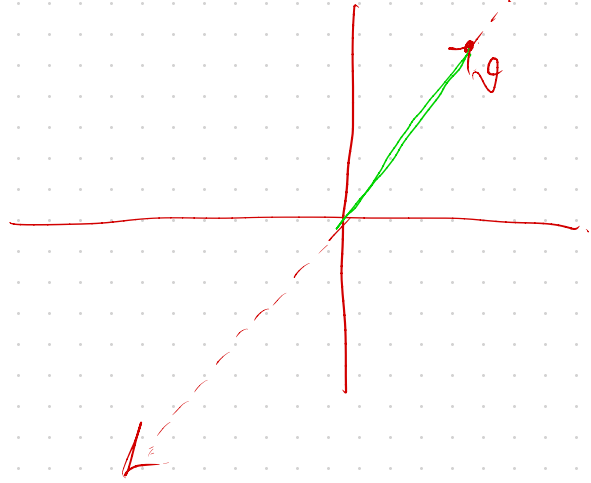
Let V and W be two vector spaces. A mapping $T : V \rightarrow W$ is called a linear transformation from V to W if it satisfies the following properties:

- ① $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$.
- ② $T(c\mathbf{v}) = cT(\mathbf{v})$ for all $\mathbf{v} \in V$ and all $c \in \mathbb{R}$.

Remark

The properties listed above in the definition are called the linearity properties. The vector space V is called the domain of T , whereas the vector space W is called the codomain of T .

$$T(cv) = cT(v)^{\uparrow}$$



V

lines $\xrightarrow{T}$ lines
in V in W

W

Example and Non-Example

$$v = WT \left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} + \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \right) = T \begin{pmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \end{pmatrix}$$

Example (A Linear Transformation)

Define $T : \mathcal{M}_2(\mathbb{R}) \rightarrow \mathcal{M}_2(\mathbb{R})$ by

$$v = T \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) = \begin{bmatrix} a - 2b & 0 \\ 3a + 4d & a + b - c \end{bmatrix}.$$

Then T is a linear transformation.

$$= T \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} + T \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} = T \begin{pmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \end{pmatrix}$$

Not a Linear Transformation

Define $T : \mathcal{M}_2(\mathbb{R}) \rightarrow \mathbb{R}$ by

$$T(A) = \det(A) \text{ for all } A \in \mathcal{M}_2(\mathbb{R}).$$

$$\begin{aligned} & T \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \\ &= T \begin{pmatrix} ka_1 & kb_1 \\ kc_1 & kd_1 \end{pmatrix} \\ &= k \det \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \end{aligned}$$

$$T \left(\underbrace{\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}}_c + \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \right) = T \left(\begin{pmatrix} \underbrace{a_1+a_2}_a & \underbrace{b_1+b_2}_b \\ \underbrace{c_1+c_2}_c & \underbrace{d_1+d_2}_d \end{pmatrix} \right)$$

$\cap$
 $M_2(\mathbb{R})$

$$= \begin{pmatrix} (a_1+a_2) - 2(b_1+b_2) & 0 \\ 3(a_1+a_2) + 4(d_1+d_2) & (a_1+a_2) + (b_1+b_2) - (c_1+c_2) \end{pmatrix}$$

||

Modified Definition of LT

Theorem (LT)

A mapping $T : V \rightarrow W$ is a linear transformation if and only if

$$T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2).$$

for all $\mathbf{v}_1, \mathbf{v}_2 \in V$ and all scalars c_1, c_2 .

Example

Let $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ be a linear transformation satisfies

$$T(1) = 2 - 3x, \quad T(x) = 2x + 5x^2, \quad T(x^2) = 3 - x + x^2.$$

For any arbitrary vector $p = a_0 + a_1x + a_2x^2$ in $\mathcal{P}_2(\mathbb{R})$, then determine $T(p)$.

Note $\{1, x, x^2\}$ is the standard basis in $\mathcal{P}_2(\mathbb{R})$

$$p(x) = a_0 + a_1x + a_2x^2; \underbrace{a_0, a_1, a_2}_{\in \mathbb{R}} \in \mathbb{R}$$

$$T(p(x)) = T(\underbrace{a_0 + a_1x + a_2x^2}_{\in \mathbb{P}_2(\mathbb{R})})$$

$$= T(a_0 + a_1x) + T(a_2x^2) \quad \left(\begin{array}{l} \text{linearity} \\ \text{of } T \end{array} \right)$$

$$= T(a_0) + T(a_1x) + T(a_2x^2)$$

$$= a_0T(1) + a_1T(x) + a_2T(x^2)$$

$$= a_0(2-3x) + a_1(2x+5x^2) + a_2(3-x+x^2)$$

$$= (2a_0 + 3a_2) + (-3a_0 + a_1 - a_2)x + (5a_1 + a_2)x^2$$

Theorem

Let $T : V \rightarrow W$ be a linear transformation. Then,

- 1 $T(\mathbf{0}_V) = \mathbf{0}_W$.
- 2 $T(-\mathbf{v}) = -1T(\mathbf{v})$ for all $\mathbf{v} \in V$.

Example

Example

Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ as follows: If $\mathbf{x} = (x_1, x_2, x_3)$, then

$$T(\mathbf{x}) = (2x_1 + x_2, x_1 + x_2 - x_3).$$

Verify that T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

$$\mathbf{y} = (y_1, y_2, y_3), \quad \mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3.$$

$$\begin{aligned} T(a\mathbf{x} + b\mathbf{y}) &= T(a(x_1, x_2, x_3) + b(y_1, y_2, y_3)) \\ (a, b \in \mathbb{R}) \quad &= T(\underbrace{(ax_1 + by_1)}_{\text{new } x_1}, \underbrace{ax_2 + by_2}_{\text{new } x_2}, \underbrace{ax_3 + by_3}_{\text{new } x_3}) \end{aligned}$$

$$x'_1 = ax_1 + by_1, \quad x'_2 = ax_2 + by_2 \quad \text{and} \quad x'_3 = ax_3 + by_3$$

By the definition,

$$T(ax_1 + by_1, ax_2 + by_2, ax_3 + by_3) = T(x'_1, x'_2, x'_3) \\ = (2x'_1 + x'_2, x'_1 + x'_2 - x'_3)$$

$$= (2(ax_1 + by_1) + (ax_2 + by_2), (ax_1 + by_1) + (ax_2 + by_2) - (ax_3 + by_3))$$

$$\left[\begin{array}{l} (x+y, z+w) \\ = (x, z) + (y, w) \\ \text{in } \mathbb{R}^2 \end{array} \right] = (2ax_1 + ax_2, ax_1 + ax_2 - ax_3) + (2by_1 + by_2, by_1 + by_2 - by_3) \\ = a(2x_1 + x_2, x_1 + x_2 - x_3) + b(2y_1 + y_2, y_1 + y_2 - y_3) \\ = aT(x) + bT(y) \quad \forall a, b \in \mathbb{R} \text{ and } x, y \in \mathbb{R}^3$$

Ex $T: \mathbb{R}^3 \rightarrow \mathbb{R}^4$

$$T(x_1, x_2, x_3) = (2x_1 + x_2, x_1 + x_2 - x_3, x_1 + x_3, 5)$$

$$(0, 0, 0) \mapsto (0, 0, 0, 5) \neq (0, 0, 0, 0)$$

T is NOT a linear transⁿ.

$$\begin{pmatrix} \begin{array}{|c|c|c|} \hline - & & \\ \hline & & \\ \hline \end{array} \\ 2 \times 3 \end{pmatrix} \begin{pmatrix} \begin{array}{|c|} \hline - \\ \hline - \\ \hline - \\ \hline \end{array} \\ 3 \times 1 \end{pmatrix} = \begin{pmatrix} \begin{array}{|c|} \hline - \\ \hline - \\ \hline \end{array} \\ 2 \times 1 \end{pmatrix}$$

(for the
previous
example
 $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$)

$$\begin{pmatrix} \begin{array}{|c|} \hline \\ \hline \\ \hline \\ \hline \end{array} \\ m \times n \end{pmatrix} \begin{pmatrix} \begin{array}{|c|} \hline \\ \hline \\ \hline \\ \hline \end{array} \\ n \times 1 \end{pmatrix} = \begin{pmatrix} \begin{array}{|c|} \hline \\ \hline \\ \hline \\ \hline \end{array} \\ m \times 1 \end{pmatrix}$$

$\mathbb{R}^n \longrightarrow \mathbb{R}^m$

Arise of LT

The next theorem gives us the idea about how linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$ arise.

Theorem

Let A be an $m \times n$ matrix, and define $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $T(\mathbf{x}) = A\mathbf{x}$. Then, T is a linear transformation.

Handwritten diagram illustrating the mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$. A red box highlights the mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Below it, a red arrow points from a vector $\mathbf{x}$ in $\mathbb{R}^n$ to its image $A\mathbf{x}$ in $\mathbb{R}^m$.

Remark

A linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined as $T(\mathbf{x}) = A\mathbf{x}$, where A is an $m \times n$ matrix, is called a matrix transformation.

Example of Matrix Transformation

Example (MT)

Determine the matrix transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ if

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix}.$$

Handwritten note: A red arrow points from the $\mathbb{R}^3$ in the text above to the matrix A . To the right of the matrix, "2 x 3" is written in red and underlined with a red squiggle.

$$(x_1, x_2, x_3) \in \mathbb{R}^3.$$

$$T(x) = Ax = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_1 + x_2 \\ x_1 + x_2 - x_3 \end{pmatrix}.$$

$$T(x_1, x_2, x_3) = (2x_1 + x_2, x_1 + x_2 - x_3).$$

$$\text{where } (x_1, x_2, x_3) \in \mathbb{R}^3.$$

$$\text{Ex: } T(x_1, x_2, x_3) = (2x_1 + x_2, x_1 + x_2 - x_3)$$

$$\mathbb{R}^3 \longrightarrow \mathbb{R}^2.$$

$$\begin{array}{ccc} & e_1, e_2, e_3 & \\ \swarrow & \downarrow & \searrow \\ (1, 0, 0) & (0, 1, 0) & (0, 0, 1) \end{array}$$

$$T(1, 0, 0) = (2, 1)$$

$$T(0, 1, 0) = (1, 1)$$

$$T(0, 0, 1) = (0, -1)$$

$$\begin{aligned} A &= [T(e_1) \ T(e_2) \ T(e_3)] \\ &= \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix} \quad (\text{Ans}) \end{aligned}$$

✓

Question

Can we always describe a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ as a matrix transformation?

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is described by the matrix transformation

$$T(\mathbf{x}) = A\mathbf{x},$$

where A is the $m \times n$ matrix

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix}$$

and $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ denote the standard basis vectors in $\mathbb{R}^n$.

$$\mathbf{e}_1 = (1, 0, \dots, 0)$$

$$\mathbf{e}_2 = (0, 1, 0, \dots, 0)$$

$$\mathbf{e}_n = (0, 0, \dots, 1, 0)$$

Matrix of T

Definition (Matrix of T)

If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, then the $m \times n$ matrix

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n)]$$

is called the matrix of T .

Example (Matrix of T)

Determine the matrix of the linear transformation $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3, x_4) = (2x_1 + 3x_2 + x_4, 5x_1 + 9x_3 - x_4, 4x_1 + 2x_2 - x_3 + 7x_4).$$

$$\begin{aligned} \mathbf{e}_1 &= (1, 0, 0, 0) & ; & \quad T(1, 0, 0, 0) = (2, 5, 4) \\ \mathbf{e}_2 &= (0, 1, 0, 0) & ; & \quad T(0, 1, 0, 0) = (3, 0, 2) \\ \mathbf{e}_3 &= (0, 0, 1, 0) & ; & \quad T(0, 0, 1, 0) = (0, 9, -1) \\ \mathbf{e}_4 &= (0, 0, 0, 1) & ; & \quad T(0, 0, 0, 1) = (1, -1, 7) \end{aligned}$$
$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad T(\mathbf{e}_3) \quad T(\mathbf{e}_4)] = \begin{pmatrix} 2 & 3 & 0 & 1 \\ 5 & 0 & 9 & -1 \\ 4 & 2 & -1 & 7 \end{pmatrix}$$

Linear Transformation Defined on the Standard Basis

Example

Determine the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ satisfying

$$T(1, 0) = (2, 3, -1), \quad T(0, 1) = (5, -4, 7).$$

Solⁿ: AIM: $T(x_1, x_2) = ?$ $x_1, x_2 \in \mathbb{R}$

$$(x_1, x_2) = x_1(1, 0) + x_2(0, 1)$$

$$T(x_1, x_2) = T(x_1(1, 0) + x_2(0, 1))$$

$$= x_1 T(1, 0) + x_2 T(0, 1)$$

$$= x_1(2, 3, -1) + x_2(5, -4, 7)$$

$$= (2x_1 + 5x_2, 3x_1 - 4x_2, -x_1 + 7x_2)$$

Finding the Linear Transformation

Example

Suppose $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is a linear transformation with $T(1, 0, 0) = (4, 5)$, $T(0, 1, 0) = (-1, 1)$, and $T(2, 1, -3) = (7, -1)$. Find $T(x_1, x_2, x_3)$.

$$e_1 = (1, 0, 0)$$

$$e_2 = (0, 1, 0)$$

$$u = (2, 1, -3)$$

Since they are
l.i. in $\mathbb{R}^3$. Hence

$\{e_1, e_2, u\}$ is a basis
in $\mathbb{R}^3$

$$\begin{aligned} \mathbb{R}^3(0, 0, 1) &= c_1(1, 0, 0) + c_2(0, 1, 0) + c_3(2, 1, -3) \\ &= (c_1 + 2c_3, c_2 + c_3, -3c_3) \end{aligned}$$

$$\Rightarrow -3c_3 = 1 \Rightarrow c_3 = -\frac{1}{3}$$

$$c_1 + 2c_3 = 0 \Rightarrow c_1 = -2c_3 = \frac{2}{3}$$

$$c_2 + c_3 = 0 \Rightarrow c_2 = -c_3 = \frac{1}{3}$$

$$(0, 0, 1) = \frac{1}{3}(1, 0, 0) + \frac{2}{3}(0, 1, 0) - \frac{1}{3}(2, 1, -3)$$

$$T(0, 0, 1) = \frac{1}{3}T(1, 0, 0) + \frac{2}{3}T(0, 1, 0) - \frac{1}{3}T(2, 1, -3)$$

② $T(1,0,0), T(0,1,0), T(0,0,1)$



$$(x_1, x_2, x_3) = x_1(1,0,0) + x_2(0,1,0) + x_3(0,0,1)$$

$$T(x_1, x_2, x_3) = \underline{\hspace{10cm}}$$

(Ans)

Simple Transformations

- ① $R_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. This gives $T(x, y) = (x, -y)$. Similarly, we can check for $R_y = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$. ✓
- ② The next set of transformations that we note down with matrices
- ❶ $LS_x = \begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$, $k > 0$: Linear stretch in x-direction.
 - ❷ $LS_y = \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$, $k > 0$: Linear stretch in y-direction.

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{matrix} 1 & 1 \\ 2 \end{matrix}$$

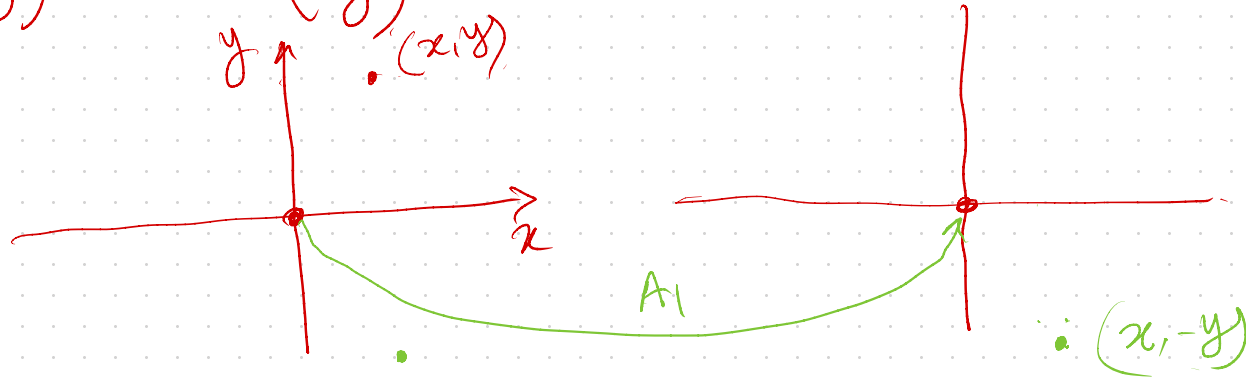
$$A_1: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

$$A_1 \approx \begin{pmatrix} \sin \theta & \cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$(x, y) \in \mathbb{R}^2$$

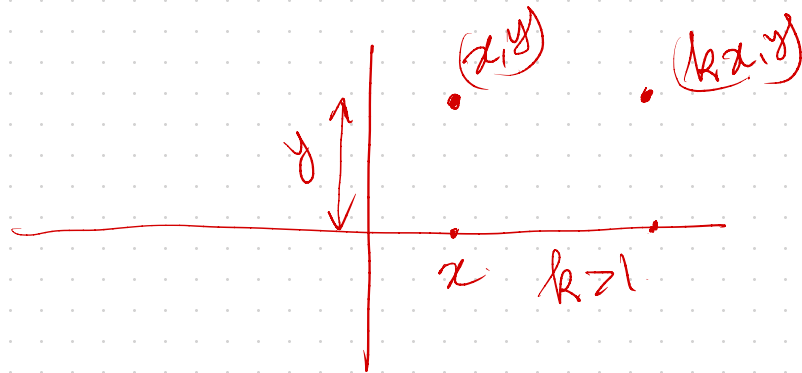
$$A_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ -y \end{pmatrix}$$

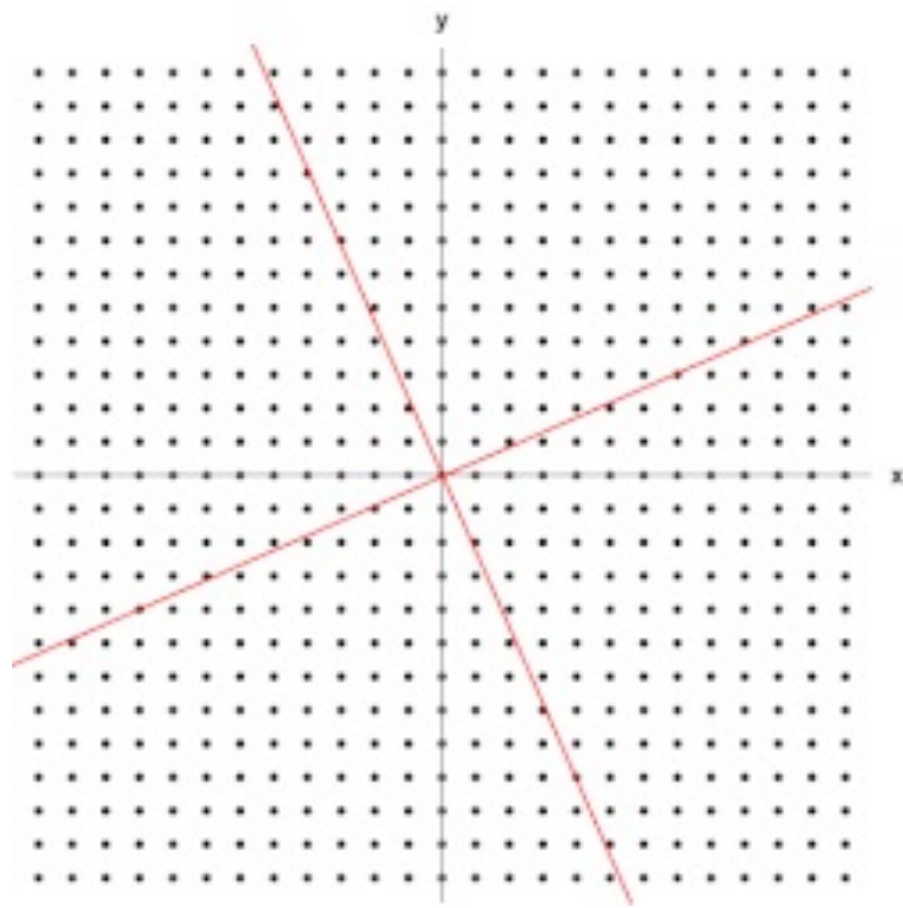


$$A_2 = \begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$$

$$A_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} kx \\ y \end{pmatrix}$$



$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$



$$A \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

eigen value.

$$\lambda \in \mathbb{R}$$

$\lambda = 0$ is allowed

$$\begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

eigen-vector.

$$Tx = \lambda x$$

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$A \rightarrow (n \times n)$$

$$Ax = \lambda x, \quad x \neq 0, \quad \lambda \in \mathbb{R}.$$

$\Rightarrow \lambda \rightarrow$ eigenvalue, $x \neq 0 \rightarrow$ eigenvector.

Thank You

Questions?